# Jonathan D. Shaw

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: https://jonathan-d-shaw.github.io


## Education

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|---|---|
| **Ph.D. in Applied Mathematics** | *Expected May 2028* |
| University of Colorado Boulder | Boulder, CO |
| **M.S. in Applied Mathematics** | *May 2026* |
| University of Colorado Boulder | Boulder, CO |
| **B.S. in Applied Mathematics** | *May 2023* |
| University of Colorado Boulder | Boulder, CO |
| *Magna cum laude* | |
| *Concentration in Aerospace Engineering* | |

## Research Interests

- Numerical partial differential equations
- Integral equations
- "Fast" algorithm design
- Multiscale modeling/simulation
- Scattering and inverse problems
- Applications of functional, asymptotic, and numerical analysis

## Publications

1. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Computationally Efficient Analytical Models of Frequency and Voltage in Low-Inertia Systems*, IEEE Transactions on Power Delivery 2026 (: arXiv)
2. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Analytical Models of Frequency and Voltage in Large-Scale All-Inverter Power Systems*, IEEE Transactions on Power Systems 2025 (: arXiv)
3. **Shaw, J. D.**; Restrepo, J. G.; Rodríguez, N., *A dynamical model of gentrification: Exploring a simple rent control strategy*, Phys. Rev. E, 2024 (: arXiv)

## Awards

- Student Travel Award, Dynamics Days US 2025

## Talks & Presentations

### Talks

- *Improving the density interpolation method*, SIAM Graduate Student Chapter Seminar, University of Colorado Boulder, January 2026 (contributed talk)
- *A dynamical system model of gentrification*, Complex/Dynamical Systems Seminar, University of Colorado Boulder, March 2025 (invited talk)
- *A dynamical system model of gentrification*, Dynamics Days US 2025, Denver, CO, January 2025 (contributed talk)

- *On the dynamics of gentrification*, Mathematical Biology Seminar, University of Colorado Boulder, October 2023 (contributed talk)

### Poster Presentations

- *A dynamical system model of gentrification*, Dynamics Days US 2023, Denver, CO, January 2023

## Research Experience

**Department of Applied Mathematics** University of Colorado Boulder
*Research Assistant* *May 2022–Present*

Quasi-direct solvers for boundary integral equations *May 2026–Present*
*Project advisor: Adrianna Gillman*

Improving the *Density Interpolation Method* *August 2025–May 2026*
*Project advisor: Eduardo Corona*

Fast solvers for power systems applications *May 2023–August 2025*
*Project advisor: Eduardo Corona*
*Project collaborators: Fiona Majeau, Bri-Mathias Hodge, Amir Sajadi*

Gentrification dynamics *May 2022–September 2024*
*Project advisors: Juan Restrepo* and *Nancy Rodríguez*

**Los Alamos National Laboratory** Los Alamos, NM
*Intern - Physics XTD-PRI* *May 2018–August 2018*

Algebraic topology for computational fluid dynamics *May 2018–August 2018*
*Project advisor: Roy S. Baty*

## Software

### Contributions

- chunkIE: A MATLAB integral equation toolbox

## Leadership & Service

**Society for Industrial and Applied Mathematics (SIAM)** University of Colorado Boulder
*President, Graduate Student Chapter* *August 2025–Present*
*Industry Liaison, Graduate Student Chapter* *August 2024–May 2025*

**Buckskin Youth Leadership Training** Richmond, VA
*Senior Staff Member* *January 2016–August 2023*